

Homework 5

Due Time: 2:00 pm, June 1 (Monday)

Problem 1. Suppose that a mathematical model fits linear programming except for the following restrictions.

1. At least one of the following two inequalities holds:

$$4x_1 - x_2 - x_3 + x_4 \leq 14,$$

$$x_1 + 2x_2 + x_3 + x_4 \leq 18.$$

2. At least two of the following three inequalities hold:

$$2x_1 + 4x_2 - x_3 + x_4 \leq 32,$$

$$-x_1 + 3x_2 + 4x_3 + x_4 \leq 42,$$

$$3x_1 - x_2 + 3x_3 - 2x_4 \leq 58.$$

Show how to reformulate these restrictions to fit an MIP model.

Solution. Let M be a sufficiently large positive constant.

1. Introduce a binary variable y_1 . The restriction can be reformulated as

$$4x_1 - x_2 - x_3 + x_4 \leq 14 + My_1,$$

$$x_1 + 2x_2 + x_3 + x_4 \leq 18 + M(1 - y_1),$$

$$y_1 \in \{0, 1\}.$$

If $y_1 = 0$, then the first inequality is enforced and the second inequality is relaxed. If $y_1 = 1$, then the second inequality is enforced and the first inequality is relaxed. Hence, at least one of the two inequalities must hold.

2. Introduce binary variables y_2, y_3, y_4 , where $y_i = 1$ means that the corresponding inequality is allowed to be violated. The restriction can be reformulated as

$$2x_1 + 4x_2 - x_3 + x_4 \leq 32 + My_2,$$

$$-x_1 + 3x_2 + 4x_3 + x_4 \leq 42 + My_3,$$

$$3x_1 - x_2 + 3x_3 - 2x_4 \leq 58 + My_4,$$

$$y_2 + y_3 + y_4 \leq 1,$$

$$y_i \in \{0, 1\}, \quad i = 2, 3, 4.$$

The constraint $y_2 + y_3 + y_4 \leq 1$ means that at most one inequality can be relaxed. Therefore, at least two of the three inequalities must hold.

□

Problem 2. Consider the following integer nonlinear programming problem:

$$\begin{aligned} &\text{Maximize} && Z = 5x_1^2 - x_1^3 + 9x_2^2 - x_2^4, \\ &\text{subject to} && x_1 + x_2 \leq 3, \\ &&& x_1, x_2 \geq 0, \\ &&& x_1, x_2 \text{ are integers.} \end{aligned}$$

This problem can be reformulated as an equivalent pure BIP problem with a linear objective function using six binary variables y_{ij} , for $i = 1, 2$ and $j = 1, 2, 3$.

1. Formulate a BIP model for this problem where the binary variables have the interpretation

$$y_{ij} = \begin{cases} 1, & \text{if } x_i = j, \\ 0, & \text{otherwise.} \end{cases}$$

2. Formulate a BIP model for this problem where the binary variables have the interpretation

$$y_{ij} = \begin{cases} 1, & \text{if } x_i \geq j, \\ 0, & \text{otherwise.} \end{cases}$$

Solution. Since $x_1 + x_2 \leq 3$ and x_1, x_2 are nonnegative integers, each variable can only take values in $\{0, 1, 2, 3\}$.

1. Suppose

$$y_{ij} = \begin{cases} 1, & \text{if } x_i = j, \\ 0, & \text{otherwise,} \end{cases} \quad i = 1, 2, \quad j = 1, 2, 3.$$

The objective function values are

$$5x_1^2 - x_1^3 = \begin{cases} 0, & x_1 = 0, \\ 4, & x_1 = 1, \\ 12, & x_1 = 2, \\ 18, & x_1 = 3, \end{cases}$$

and

$$9x_2^2 - x_2^4 = \begin{cases} 0, & x_2 = 0, \\ 8, & x_2 = 1, \\ 20, & x_2 = 2, \\ 0, & x_2 = 3. \end{cases}$$

Therefore, the BIP formulation is

$$\begin{aligned} \text{Maximize} \quad & Z = 4y_{11} + 12y_{12} + 18y_{13} + 8y_{21} + 20y_{22} \\ \text{subject to} \quad & y_{11} + y_{12} + y_{13} \leq 1, \\ & y_{21} + y_{22} + y_{23} \leq 1, \\ & y_{11} + 2y_{12} + 3y_{13} + y_{21} + 2y_{22} + 3y_{23} \leq 3, \\ & y_{ij} \in \{0, 1\}, \quad i = 1, 2, \quad j = 1, 2, 3. \end{aligned}$$

The first two constraints ensure that each x_i takes at most one value from $\{1, 2, 3\}$. If all corresponding binary variables are zero, then $x_i = 0$. The third constraint represents $x_1 + x_2 \leq 3$.

2. Suppose

$$y_{ij} = \begin{cases} 1, & \text{if } x_i \geq j, \\ 0, & \text{otherwise,} \end{cases} \quad i = 1, 2, \quad j = 1, 2, 3.$$

Then

$$x_i = y_{i1} + y_{i2} + y_{i3}.$$

The marginal objective contributions for x_1 are

$$5(1)^2 - 1^3 = 4, \quad (5(2)^2 - 2^3) - 4 = 8, \quad (5(3)^2 - 3^3) - 12 = 6.$$

The marginal objective contributions for x_2 are

$$9(1)^2 - 1^4 = 8, \quad (9(2)^2 - 2^4) - 8 = 12, \quad (9(3)^2 - 3^4) - 20 = -20.$$

Therefore, the BIP formulation is

$$\begin{aligned} \text{Maximize} \quad & Z = 4y_{11} + 8y_{12} + 6y_{13} + 8y_{21} + 12y_{22} - 20y_{23} \\ \text{subject to} \quad & y_{13} \leq y_{12} \leq y_{11}, \\ & y_{23} \leq y_{22} \leq y_{21}, \\ & y_{11} + y_{12} + y_{13} + y_{21} + y_{22} + y_{23} \leq 3, \\ & y_{ij} \in \{0, 1\}, \quad i = 1, 2, \quad j = 1, 2, 3. \end{aligned}$$

The monotonicity constraints ensure that the binary variables correctly represent the condition $x_i \geq j$. The third constraint is exactly $x_1 + x_2 \leq 3$.

□

Problem 3. Fly-Right Airplane Company has recently received purchase requests from three customers. Because of limited production capacity, the company cannot accept all requests. The relevant data are given below.

	Customer		
	1	2	3
Start-up cost	\$4 million	\$2 million	0
Marginal net revenue	\$2.2 million	\$2.8 million	\$0.9 million
Capacity used per plane	25%	35%	20%
Maximum order	3 planes	2 planes	5 planes

Formulate a mixed-integer programming model to determine how many airplanes to produce for each customer in order to maximize total profit.

Solution. Let

$$x_i = \text{number of airplanes produced for customer } i, \quad i = 1, 2, 3.$$

For customers 1 and 2, introduce binary variables

$$y_i = \begin{cases} 1, & \text{if production is initiated for customer } i, \\ 0, & \text{otherwise,} \end{cases} \quad i = 1, 2.$$

The MIP formulation is

$$\begin{aligned}
&\text{Maximize} && Z = 2.2x_1 + 2.8x_2 + 0.9x_3 - 4y_1 - 2y_2 \\
&\text{subject to} && 0.25x_1 + 0.35x_2 + 0.20x_3 \leq 1, \\
&&& x_1 \leq 3y_1, \\
&&& x_2 \leq 2y_2, \\
&&& 0 \leq x_1 \leq 3, \\
&&& 0 \leq x_2 \leq 2, \\
&&& 0 \leq x_3 \leq 5, \\
&&& x_1, x_2, x_3 \in \mathbb{Z}, \\
&&& y_1, y_2 \in \{0, 1\}.
\end{aligned}$$

The first constraint represents the total production-capacity limit. The constraints $x_1 \leq 3y_1$ and $x_2 \leq 2y_2$ ensure that the start-up cost is incurred whenever any airplanes are produced for customers 1 and 2, respectively. Since customer 3 has no start-up cost, no binary variable is needed for customer 3. $\square$

Problem 4. Four cargo ships will be used for shipping goods from one port to four other ports, labeled 1, 2, 3, 4. Any ship can be used for making any one of these four trips. The total costs are given in the following table:

Ship	Port			
	1	2	3	4
1	500	400	600	700
2	600	600	700	500
3	700	500	700	600
4	500	400	600	600

The objective is to assign the four ships to four different ports so as to minimize the total cost.

1. Formulate this problem as an integer programming model.
2. Solve this problem by considering its LP relaxation.
3. Explain why Simplex method still gives an integer optimal solution for this type of problem even after relaxing the binary restrictions.

Solution. 1. Let

$$x_{ij} = \begin{cases} 1, & \text{if ship } i \text{ is assigned to port } j, \\ 0, & \text{otherwise.} \end{cases}$$

The integer programming formulation is

$$\begin{aligned} \text{Minimize } Z = & 500x_{11} + 400x_{12} + 600x_{13} + 700x_{14} \\ & + 600x_{21} + 600x_{22} + 700x_{23} + 500x_{24} \\ & + 700x_{31} + 500x_{32} + 700x_{33} + 600x_{34} \\ & + 500x_{41} + 400x_{42} + 600x_{43} + 600x_{44} \end{aligned}$$

subject to

$$\begin{aligned} \sum_{j=1}^4 x_{ij} &= 1, & i = 1, 2, 3, 4, \\ \sum_{i=1}^4 x_{ij} &= 1, & j = 1, 2, 3, 4, \\ x_{ij} &\in \{0, 1\}, & i, j = 1, 2, 3, 4. \end{aligned}$$

The first set of constraints requires each ship to be assigned to exactly one port. The second set of constraints requires each port to receive exactly one ship.

2. We solve the LP relaxation by the transportation simplex method.

For simplicity, divide all costs by 100. Thus the cost matrix is

$$C = \begin{pmatrix} 5 & 4 & 6 & 7 \\ 6 & 6 & 7 & 5 \\ 7 & 5 & 7 & 6 \\ 5 & 4 & 6 & 6 \end{pmatrix}.$$

The final objective value should therefore be multiplied by 100.

The LP relaxation is

$$\min \sum_{i=1}^4 \sum_{j=1}^4 c_{ij} x_{ij},$$

subject to

$$\begin{aligned} \sum_{j=1}^4 x_{ij} &= 1, & i &= 1, 2, 3, 4, \\ \sum_{i=1}^4 x_{ij} &= 1, & j &= 1, 2, 3, 4, \\ x_{ij} &\geq 0, & i, j &= 1, 2, 3, 4. \end{aligned}$$

Since this is a 4×4 transportation problem, a basic feasible solution has

$$4 + 4 - 1 = 7$$

basic variables. A zero basic variable is denoted by 0_B .

Start with the following initial basic feasible solution:

	1	2	3	4	supply
1	1	0_B	.	.	1
2	.	1	0_B	.	1
3	.	.	1	0_B	1
4	.	.	.	1	1
demand	1	1	1	1	

The current objective value is

$$Z = 5 + 6 + 7 + 6 = 24.$$

For the basic variables, solve

$$u_i + v_j = c_{ij}.$$

Taking $u_1 = 0$, we obtain

$$u = (0, 2, 2, 2), \quad v = (5, 4, 5, 4).$$

The reduced costs are

$$\Delta_{ij} = c_{ij} - u_i - v_j.$$

Thus,

	1	2	3	4
1	B	B	1	3
2	-1	B	B	-1
3	0	-1	B	B
4	-2	-2	-1	B

For a minimization problem, a negative reduced cost means that the objective value can be improved. Choose x_{42} to enter the basis.

The closed path is

$$x_{42}^+, x_{22}^-, x_{23}^+, x_{33}^-, x_{34}^+, x_{44}^-.$$

Therefore,

$$\theta = \min\{x_{22}, x_{33}, x_{44}\} = \min\{1, 1, 1\} = 1.$$

Let x_{22} leave the basis. The new basic feasible solution is

	1	2	3	4	supply
1	1	0 _B	.	.	1
2	.	.	1	.	1
3	.	.	0 _B	1	1
4	.	1	.	0 _B	1
demand	1	1	1	1	

The current objective value is

$$Z = 5 + 7 + 6 + 4 = 22.$$

Now solve $u_i + v_j = c_{ij}$ for the new basis. Taking $u_1 = 0$, we get

$$u = (0, 0, 0, 0), \quad v = (5, 4, 7, 6).$$

The reduced-cost table is

	1	2	3	4
1	B	B	-1	1
2	1	2	B	-1
3	2	1	B	B
4	0	B	-1	B

Choose x_{24} to enter the basis.

The closed path is

$$x_{24}^+, x_{23}^-, x_{33}^+, x_{34}^-.$$

Therefore,

$$\theta = \min\{x_{23}, x_{34}\} = \min\{1, 1\} = 1.$$

Let x_{23} leave the basis. The new basic feasible solution is

	1	2	3	4	supply
1	1	0 _B	.	.	1
2	.	.	.	1	1
3	.	.	1	0 _B	1
4	.	1	.	0 _B	1
demand	1	1	1	1	

The current objective value is

$$Z = 5 + 5 + 7 + 4 = 21.$$

Again solve $u_i + v_j = c_{ij}$. Taking $u_1 = 0$, we obtain

$$u = (0, -1, 0, 0), \quad v = (5, 4, 7, 6).$$

The reduced-cost table is

	1	2	3	4
1	B	B	-1	1
2	2	3	1	B
3	2	1	B	B
4	0	B	-1	B

There are still negative reduced costs. Choose x_{43} to enter the basis.

The closed path is

$$x_{43}^+, x_{44}^-, x_{34}^+, x_{33}^-.$$

Therefore,

$$\theta = \min\{x_{44}, x_{33}\} = \min\{0, 1\} = 0.$$

This is a degenerate pivot. Let x_{44} leave the basis. The solution does not change, but the basis changes to

	1	2	3	4	supply
1	1	0 _B	.	.	1
2	.	.	.	1	1
3	.	.	1	0 _B	1
4	.	1	0 _B	.	1
demand	1	1	1	1	

Now solve $u_i + v_j = c_{ij}$ one more time. Taking $u_1 = 0$, we obtain

$$u = (0, 0, 1, 0), \quad v = (5, 4, 6, 5).$$

The reduced-cost table is

	1	2	3	4
1	B	B	0	2
2	1	2	1	B
3	1	0	B	B
4	0	B	B	1

All reduced costs are nonnegative, so the current solution is optimal.

Thus an optimal solution is

$$x_{11} = 1, \quad x_{24} = 1, \quad x_{33} = 1, \quad x_{42} = 1,$$

and all other $x_{ij} = 0$. Therefore, one optimal assignment is

$$\text{Ship 1} \rightarrow \text{Port 1}, \quad \text{Ship 2} \rightarrow \text{Port 4},$$

$$\text{Ship 3} \rightarrow \text{Port 3}, \quad \text{Ship 4} \rightarrow \text{Port 2}.$$

The minimum cost is

$$Z^* = 21 \times 100 = 2100.$$

3. The reason is that every corner point of this feasible region is integer.

To see this, suppose that a feasible solution contains a fractional positive variable x_{ij} . Since the sum of every row is equal to 1, row i must contain another positive variable. Similarly, since the sum of every column is equal to 1, column j must also contain another positive

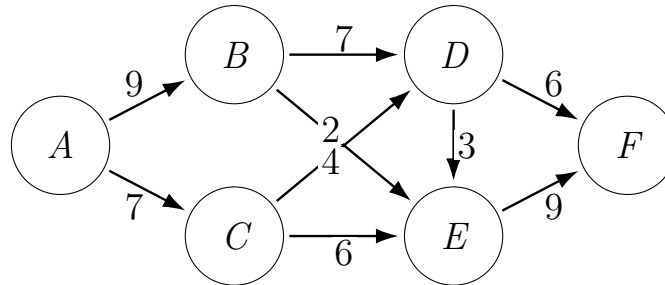
variable. Continuing this argument, we can find a cycle of positive variables in the assignment table.

Along this cycle, we can alternately add and subtract a small number $\epsilon > 0$ from the variables on the cycle. If ϵ is chosen sufficiently small, all variables remain nonnegative, and every row sum and column sum remains equal to 1. Therefore, the original fractional solution can be written as the average of two different feasible solutions. Hence it is not a corner point.

Thus, no fractional feasible solution can be a corner point. Therefore, every corner point of the LP relaxation is a 0-1 solution. Since the simplex method always finds an optimal solution at a corner point when an optimum exists, solving the LP relaxation still gives an integer optimal assignment.

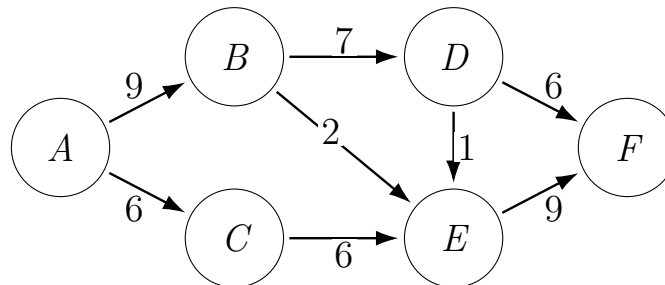
□

Problem 5. Consider the maximum flow problem shown below. The source is node A , the sink is node F , and the arc capacities are the numbers shown next to the directed arcs.



1. Use the augmenting path method to solve this maximum flow problem. List each augmenting path, the amount of flow augmented along that path, and the final flow on each arc.
2. Formulate this maximum flow problem as a linear programming problem. Define the decision variables, objective function, capacity constraints, and flow conservation constraints explicitly.
3. Prove that the solution obtained in part (1) is optimal.

Solution. 1. The max flow of the given problem.



2. Let x_{ij} denote the amount of flow sent on arc (i, j) , and let v denote the total flow value from source A to sink F . The maximum flow problem can be formulated as follows:

$$\max \quad v$$

subject to

$$x_{AB} + x_{AC} = v,$$

$$x_{DF} + x_{EF} = v,$$

and the flow conservation constraints at the intermediate nodes:

$$x_{AB} = x_{BD} + x_{BE},$$

$$x_{AC} = x_{CD} + x_{CE},$$

$$x_{BD} + x_{CD} = x_{DE} + x_{DF},$$

$$x_{BE} + x_{CE} + x_{DE} = x_{EF}.$$

The capacity constraints are

$$0 \leq x_{AB} \leq 9, \quad 0 \leq x_{AC} \leq 7,$$

$$0 \leq x_{BD} \leq 7, \quad 0 \leq x_{BE} \leq 2,$$

$$0 \leq x_{CD} \leq 4, \quad 0 \leq x_{CE} \leq 6,$$

$$0 \leq x_{DE} \leq 3, \quad 0 \leq x_{DF} \leq 6, \quad 0 \leq x_{EF} \leq 9.$$

3. Consider the cut

$$S = \{A, B, C, D, E\}, \quad T = \{F\}.$$

The only arcs going from S to T are

$$(D, F) \quad \text{and} \quad (E, F).$$

Therefore, the capacity of this cut is

$$u_{DF} + u_{EF} = 6 + 9 = 15.$$

By the duality theory of linear programming, the dual of the maximum flow LP is the minimum cut LP. Equivalently, every feasible cut gives a feasible solution to the dual problem, and its cut capacity gives the corresponding dual objective value. Since the cut above has capacity 15, it gives a dual feasible solution with objective value 15.

The feasible flow from part (1) has primal objective value 15, while the cut above gives a dual feasible solution with objective value 15. By weak duality, no feasible flow can have value larger than the value of any feasible cut. Hence,

$$\text{maximum flow value} \leq 15.$$

Since we already have a feasible flow of value 15, it follows that

$$\text{maximum flow value} = 15.$$

Therefore, the solution obtained in part (1) is optimal.

□